



NATIONAL SENIOR CERTIFICATE EXAMINATION
NOVEMBER 2013

INFORMATION TECHNOLOGY: PAPER II

MARKING GUIDELINES

Time: 3 hours

120 marks

These marking guidelines are prepared for use by examiners and sub-examiners, all of whom are required to attend a standardisation meeting to ensure that the guidelines are consistently interpreted and applied in the marking of candidates' scripts.

The IEB will not enter into any discussions or correspondence about any marking guidelines. It is acknowledged that there may be different views about some matters of emphasis or detail in the guidelines. It is also recognised that, without the benefit of attendance at a standardisation meeting, there may be different interpretations of the application of the marking guidelines.

SECTION A SQL ANSWER SHEET

1.1:

```
SELECT *FROM tblShows ORDER BY ShowDate
```

1.2:

```
SELECT ShowName, ShowTime, ShowCapacity FROM tblShows  
WHERE ShowTime = #15:00# AND ShowCapacity >= 250;
```

1.3:

```
SELECT ActorName, INT ((NOW() - ActorDOB) / 365) AS ActorAge FROM tblActors;
```

OR

```
SELECT ActorName, YEAR (NOW()) – YEAR(ActorDOB) AS ActorAge FROM tblActors;
```

1.4:

```
SELECT ShowName, COUNT(RoleName)  
FROM tblRoles , tblShows  
WHERE TblRoles.ShowID = tblShows.ShowID  
GROUP BY ShowName
```

1.5:

```
INSERT INTO tblActors (ActorName, ActorDOB, ActorRate)  
VALUES ('Bob Reeves', #1976-10-04#, 175)
```

1.6:

```
UPDATE tblShows SET ShowCapacity = 450  
WHERE ShowID = "Henry5";
```

1.7:

```
SELECT ShowDate, SUM(ShowCapacity * ShowCost) AS TotalIncome  
FROM tblShows GROUP BY ShowDate;
```

1.8:

```
SELECT ShowName, RoleName  
FROM tblRoles, tblShows  
WHERE tblRoles.ShowID = tblShows.ShowID AND RoleName LIKE 'Sir*'
```

1.9:

```
SELECT ActorName, ShowName, ROUND(SUM(ShowDuration / 60 * ActorRate),2)  
FROM tblShows, tblActors, tblRoles  
WHERE tblShows.ShowID = tblRoles.ShowID AND tblRoles.ActorID = tblActors.ActorID  
GROUP BY ActorName, ShowName;
```

SECTION B OBJECT ORIENTED PROGRAMMING

JAVA SOLUTION:

QUESTION 2.

```
// Question 2.1
public class Booking
{
    // Question 2.2
    private String seat;
    private boolean paid;
    private String contact;
    // Question 2.3
    public Booking(String st, String ct, boolean pd)
    {
        seat = st;
        paid = pd;
        contact = ct;
    }

    // Question 2.4
    public double getCost()
    {
        double cost = 200.00;

        if (seat.charAt(0) >= 'E' & seat.charAt(0) <= 'H')
        {
            cost = cost * 0.8;
        }
        else if (seat.charAt(0) >= 'I' & seat.charAt(0) <= 'L')
        {
            cost = cost * 0.6;
        }

        return cost;
    }

    // Question 2.5
    public String getSeat()
    {
        return seat;
    }

    public boolean isPaid()
    {
        return paid;
    }

    // Question 2.6
    public String toString()
    {
        String tmp = getSeat() + "\tR" + Math.Round(getCost()*100)/100.0 +
                     "\t\t";

        if (isPaid())
        {
            tmp = tmp + "Pd";
        }

        return tmp;
    }
}
```

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QUESTION 3

```
// Question 3.1
public class GroupBooking extends Booking
{
    // Question 3.2
    private String groupName;
    private int groupSize;
    // Question 3.3
    public GroupBooking(String st, String ct, boolean pd, String gn, int gs)
    {
        super(st,ct,pd);
        groupName = gn;
        groupSize = gs;
    }
    // Question 3.4
    public double getCost()
    {
        double cost = super.getCost();
        int discountPoints = groupSize / 4;

        return cost - (20.00 * discountPoints);
    }
    // Question 3.5
    public String getGroupName()
    {
        return groupName;
    }
    // Question 3.6
    public String toString()
    {
        return super.toString() + "\t" + groupName;
    }
}
```

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QUESTIONS 4 AND 6.1

```
import java.io.*;

// Question 4.1
public class BookingManager
{
    // Question 4.2
    private Booking[] bookings = new Booking[180];
    private int counter = 0;
}
```

Private
Types (Booking array and int)
Initialised correctly

```
// Question 4.3
public BookingManager()
{
    try
    {
        BufferedReader fIn = new BufferedReader(new
            FileReader("bookings.txt"));

        String line = fIn.readLine();

        while (line != null)
        {
            String[] parts = line.split(",");

            boolean paid = (parts[2].equals("Yes"));

            if (parts.length == 3)
            {
                bookings[counter] = new Booking(parts[0], parts[1], paid);
            }
            else
            {
                int groupSize = Integer.parseInt(parts[4]);

                bookings[counter] = new GroupBooking(parts[0], parts[1],
                    paid, parts[3], groupSize);
            }

            counter++;
            line = fIn.readLine();
        }

        fIn.close();
    }
    catch (IOException e)
    {
        System.out.println(e);
    }
}

// Question 4.4
public String listAllBookings()
{
    String rString = "";

    for(int i = 0; i < counter; i++)
    {
        rString += bookings[i] + "\n";
    }

    return rString;
}
```

```
// Question 4.5
public String listGroupBookings()
{
    String rString = "";

    for(int i = 0; i < counter; i++)
    {
        if (bookings[i] instanceof GroupBooking)
        {
            GroupBooking tempBooking = (GroupBooking) bookings[i];

            if (rString.indexOf(tempBooking.getGroupName()) < 0)
            {
                rString += tempBooking.getGroupName() + "\n";
            }
        }
    }

    return rString;
}
```

```
// Question 4.6
public double outstandingPayments()
{
    double totalOutstanding = 0.00;

    for(int i = 0; i < counter; i++)
    {
        if (!bookings[i].isPaid())
        {
            totalOutstanding += bookings[i].getCost();
        }
    }

    return totalOutstanding;
}
```

[30]

```
// Question 6.1 (Marked with Rubric)
public String listEmptySeats()
{
    String emptySeats = "";

    for (char c = 'A'; c <= 'L' c++)
    {
        for (int j = 1; j <= 15; j++)
        {
            String currentSeat = "" + c + j;
            int seatCounter = 0;
            boolean foundSeat = false;

            while ((seatCounter < counter) && (foundSeat == false))
            {
                if (bookings[seatCounter].getSeat().equals(currentSeat))
                {
                    foundSeat = true;
                }
                seatCounter++;
            }

            if (!foundSeat)
            {
                emptySeats += currentSeat + " ";
            }
        }
    }
}
```

```

        }
    }
    return emptySeats;
}
}

```

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QUESTIONS 5 AND 6.2

```

import java.io.BufferedReader;
import java.io.InputStreamReader;

// Question 5.1
public class TheatreManager
{
    public static void main(String[] args) throws Exception
    {
        BookingManager bMan = new BookingManager();

        // Question 5.2
        System.out.println("All Bookings");
        System.out.println("-----");
        System.out.println(bMan.listAllBookings());
        System.out.println();

        System.out.println("Group Bookings");
        System.out.println("-----");
        System.out.println(bMan.listGroupBookings());
        System.out.println();

        System.out.println("Outstanding Payments");
        System.out.println("-----");
        System.out.println("Outstanding Payments : R " +
                           bMan.outstandingPayments());
        System.out.println();

        // Question 6.2
        System.out.println("Empty Seats");
        System.out.println("-----");
        System.out.println(bMan.listEmptySeats());
    }
}

```

Call Methods
Correct Order
Correct Output

[5]

[1]

DELPHI SOLUTION:

QUESTION 2

```
unit uBooking;

interface

uses SysUtils;

// Question 2.1
type TBooking = class
    private // Question 2.2
        seat : String;
        paid : boolean;
        contact : String;
    public
        constructor Create(st : String; ct : String; pd : boolean);
        function getCost : real; virtual;
        function getSeat : String;
        function isPaid : boolean;
        function toString : String; virtual;
end;

implementation

{ TBooking }

// Question 2.3
constructor TBooking.Create(st: String; ct: String ; pd: boolean);
begin
    seat := st;
    paid := pd;
    contact := ct;
end;

// Question 2.4
function TBooking.getCost: real;
var
    cost : real;
begin
    cost := 200.00;

    if (seat[1] >= 'E') AND (seat[1] <= 'H') then
        begin
            cost := cost * 0.8;
        end
    else if (seat[1] >= 'I') AND (seat[1] <= 'L') then
        begin
            cost := cost * 0.6;
        end;

    Result := cost;
end;

// Question 2.5
function TBooking.getSeat: String;
begin
    Result := seat;
end;

// Question 2.6
function TBooking.isPaid: boolean;
begin
```

```
    Result := paid;
end;

function TBooking.toString: String;
begin
    Result := getSeat + #9 + 'R' + FloatToStrF(getCost, ffFixed, 10, 2) + #9 +
              #9;

    if isPaid then
        Result := Result + 'Pd';
    end;
end.
```

[17]

QUESTION 3

```
unit uGroupBooking;

interface

uses uBooking;

// -- Question 3.1 --

type TGroupBooking = class (TBooking)

// -- Question 3.2 --

private
    groupName : String;
    groupSize : integer;
public
    constructor Create (st : String; ct : String; pd : boolean; gn : String; gs
                        : integer);
    function getCost : real; override;
    function toString : String; override;
    function getGroupName : String;
end;

implementation

{ TGroupBooking }

// -- Question 3.3 --

constructor TGroupBooking.Create(st: String; ct: String; pd: boolean; gn:
                                String;
                                gs: integer);
begin
    Inherited Create(st,ct,pd);
    groupName := gn;
    groupSize := gs;
end;
```

```
// -- Question 3.4 --

function TGroupBooking.getCost: real;
var
    cost : real;
    discountPoints : integer;
begin
    cost := Inherited getCost;
    discountPoints := groupSize DIV 4;

    Result := cost - (20.00 * discountPoints);
end;

// -- Question 3.5 --

function TGroupBooking.getGroupName: String;
begin
    Result := groupName;
end;

// -- Question 3.6 --

function TGroupBooking.toString: String;
begin
    Result := Inherited toString + #9 + groupName;
end;

end.
```

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QUESTIONS 4 AND 6.1

```
unit uBookingManager;

interface

uses uBooking, uGroupBooking, SysUtils;

// -- Question 4.1 -

type TBookingManager = class

// -- Question 4.2 --

    private
        bookings : array[1..180] of TBooking;
        counter : integer;
    public
        constructor Create;
        function listAllBookings : String;
        function listGroupBookings : String;
        function outstandingPayments : real;
        function listEmptySeats : String;
end;

implementation

{ TBookingManager }
// -- Question 4.3 --

constructor TBookingManager.Create;
var
```

} Private
Types (Booking array and int)
Initialised correctly

```
inFile : TextFile;
line : String;
seat, contact, gName, sPaid : String;
gSize : integer;
paid : boolean;
begin

AssignFile(inFile, 'bookings.txt');
Reset(inFile);

counter := 0;

while not EOF(infile) do
begin
    ReadLn(inFile, line);
    Inc(counter);

    seat := Copy(line, 1, Pos(',', line) - 1);
    Delete(line, 1, Pos(',', line));

    contact := Copy(line, 1, Pos(',', line) - 1);
    Delete(line, 1, Pos(',', line));

    if Pos(',', line) > 0 then
begin
        // GroupBooking
        sPaid := Copy(line, 1, Pos(',', line) - 1);
        Delete(line, 1, Pos(',', line));

        paid := (sPaid = 'Yes');

        gName := Copy(line, 1, Pos(',', line) - 1);
        Delete(line, 1, Pos(',', line));

        gSize := StrToInt(line);

        bookings[counter] := TGroupBooking.Create(seat, contact, paid, gName,
                                                    gSize);
    end
else
begin
        // Booking
        sPaid := line;

        paid := (sPaid = 'Yes');

        bookings[counter] := TBooking.Create(seat, contact, paid);
    end;
end;

CloseFile(inFile);

end;
```

```
// -- Question 4.4 --
function TBookingManager.listAllBookings: String;
var
    loop : integer;
begin
    for loop := 1 to counter do
        Result := Result + bookings[loop].toString + #13;
    end;

// -- Question 4.5 --
function TBookingManager.listGroupBookings: String;
var
    temp : String;
    loop : integer;
    tempBooking : TGroupBooking;
begin
    temp := '';

    for loop := 1 to counter do
        begin
            if bookings[loop] is TGroupBooking then
                begin
                    tempBooking := bookings[loop] as TGroupBooking;
                    if (Pos(tempBooking.getGroupName, temp) <= 0) then
                        begin
                            temp := temp + tempBooking.getGroupName + #13;
                        end;
                    end;
                end;
            end;

        Result := temp;

    end;

// -- Question 4.6 --
function TBookingManager.outstandingPayments: real;
var
    totalOutstanding : real;
    loop : integer;
begin
    totalOutstanding := 0;

    for loop := 1 to counter do
        if bookings[loop].isPaid = false then
            totalOutstanding := totalOutstanding + bookings[loop].getCost;

    Result := totalOutstanding;
end;
```

[30]

```
// -- Question 6.1 -- (Marked with Rubric)
function TBookingManager.listEmptySeats: String;
var
    rowCounter : char;
    numCounter : integer;
    emptySeats : String;
    foundSeat : boolean;
    currentSeat : String;
    seatCounter : integer;
begin
    emptySeats := '';

    for rowCounter := 'A' to 'L' do
        begin
            for numCounter := 1 to 15 do
                begin
                    currentSeat := rowCounter + IntToStr(numCounter);
                    seatCounter := 1;
                    foundSeat := false;
                    while (seatCounter <= counter) AND (foundSeat = false) do
                        begin
                            if bookings[seatCounter].getSeat = currentSeat then
                                foundSeat := true;
                                Inc(seatCounter);
                            end;

                            if foundSeat = false then
                                emptySeats := emptySeats + currentSeat + ' ';
                            end;
                        end;
                    end;

                    Result := emptySeats;
                end;
            end;
        end;
    end.
```

[11]

QUESTIONS 5 AND 6.2

```
unit uFrmMain;
interface

uses
  Windows, Messages, SysUtils, Variants, Classes, Graphics, Controls, Forms,
  Dialogs, uBookingManager, StdCtrls, ComCtrls;

// -- Question 5.1 --
type
  TTheatreManager = class(TForm)
    rchOutput: TRichEdit;
    procedure FormCreate(Sender: TObject);
    procedure FormActivate(Sender: TObject);
  private
    { Private declarations }
  public
    { Public declarations }
  end;
var
  Form1: TTheatreManager;
  bMan : TBookingManager;

implementation
{$R *.dfm}

procedure TForm1.FormCreate(Sender: TObject);
begin
  bMan := TBookingManager.Create;
end;

procedure TForm1.FormActivate(Sender: TObject);
begin
  // -- Question 5.2 --
  rchOutput.Lines.Add('All Bookings');
  rchOutput.Lines.Add('-----');
  rchOutput.Lines.Add(bMan.listAllBookings);
  rchOutput.Lines.Add('');

  rchOutput.Lines.Add('Group Bookings');
  rchOutput.Lines.Add('-----');
  rchOutput.Lines.Add(bMan.listGroupBookings);
  rchOutput.Lines.Add('');

  rchOutput.Lines.Add('Outstanding Payments');
  rchOutput.Lines.Add('-----');
  rchOutput.Lines.Add('Outstanding Payments : R' +
    FloatToStr(bMan.outstandingPayments));
  rchOutput.Lines.Add('');

  // -- Question 6.2 --
  rchOutput.Lines.Add('Empty Seats');
  rchOutput.Lines.Add('-----');
  rchOutput.Lines.Add(bMan.listEmptySeats);
end;

end.
```

Call Methods
Correct Order
Correct Output

[5]

[1]

OUTPUT

SECTION A STRUCTURED QUERY LANGUAGE

QUESTION 1.1

Question_1_1						
ShowID	ShowName	ShowDate	ShowTime	ShowDuration	ShowCapacity	ShowCost
Ado	Much Ado about Nothing	2013/11/03	15:00	109	415	200
Lear	King Lear	2013/11/04	17:30	179	350	250
12Night	Twelfth Night	2013/11/08	15:00	160	325	200
Macbeth	Macbeth	2013/11/09	17:30	174	280	250
Henry5	Henry V	2013/11/12	17:30	270	300	250
Othello	Othello	2013/11/20	20:00	190	300	350
Merchant	Merchant of Venice	2013/11/20	17:30	279	400	250
2Gents	Two Gentlemen of Verona	2013/11/20	15:00	215	350	200
Romeo	Romeo and Juliet	2013/11/22	17:30	160	360	250
MND	Midsummer Night's Dream	2013/11/24	15:00	204	375	200
Errors	Comedy of Errors	2013/11/24	20:00	155	300	350

QUESTION 1.2

Question_1_2		
ShowName	ShowTime	ShowCapacity
Twelfth Night	15:00	325
Two Gentlemen of Verona	15:00	350
Much Ado about Nothing	15:00	415
Midsummer Night's Dream	15:00	375

QUESTION 1.3

Question_1_3	
ActorName	ActorAge
Garth York	33
Ebony Sweeney	32
Mary Rocha	51
Summer Dickson	43
Clarke Stafford	44
Veronica Decker	30
Jason Hyde	50
Odette Barker	31
Oprah Buckner	31
Amir Collier	48
Cade Ewing	34
Callie Moon	30
Britanni Puckett	32

Question_1_3	
ActorName	ActorAge
Zelda Fry	50
Maggy House	35
Elliott Saunders	49
Noelle Dillard	38
Donna Mullen	47
Leonard Hickman	30
Debra David	37
Denton French	36
Darius Becker	48
Palmer Reyes	44
Jared Larson	40
Prescott McMahon	45
Chadwick Gardner	47
Galvin Merritt	32
Dacey Mathis	40
Hall Holt	50
Willow Woods	32

QUESTION 1.4

Question_1_4	
ShowName	TotalRoles
Comedy of Errors	16
Henry V	27
King Lear	17
Macbeth	30
Merchant of Venice	17
Midsummer Night's Dream	21
Much Ado about Nothing	18
Othello	15
Romeo and Juliet	23
Twelfth Night	15
Two Gentlemen of Verona	15

QUESTION 1.5

(No output)

QUESTION 1.6

(No output)

QUESTION 1.7

Question_1_7	
ShowDate	TotalIncome
2013/11/03	83000
2013/11/04	87500
2013/11/08	65000
2013/11/09	70000
2013/11/12	75000
2013/11/20	275000
2013/11/22	90000
2013/11/24	180000

QUESTION 1.8

Question_1_8	
ShowName	RoleName
Twelfth Night	Sir Andrew Aguecheek
Henry V	Sir Thomas Erpingham
Henry V	Sir Thomas Grey
Twelfth Night	Sir Toby Belch

QUESTION 1.9

Query1_9		
ActorName	ShowName	Expr1002
Amir Collier	Henry V	3375
Amir Collier	King Lear	2237.5
Amir Collier	Macbeth	2175
Amir Collier	Merchant of Venice	3487.5
Amir Collier	Midsummer Night's Dream	2550
Amir Collier	Romeo and Juliet	2000
Amir Collier	Two Gentlemen of Verona	2687.5
Britanni Puckett	Comedy of Errors	1937.5
Britanni Puckett	Henry V	3375
Britanni Puckett	King Lear	2237.5
Britanni Puckett	Macbeth	2175
Britanni Puckett	Midsummer Night's Dream	2550
Britanni Puckett	Othello	2375
Britanni Puckett	Romeo and Juliet	2000
Britanni Puckett	Twelfth Night	2000
Britanni Puckett	Two Gentlemen of Verona	2687.5
Cade Ewing	Henry V	1237.5

Query1_9		
ActorName	ShowName	Expr1002
Cade Ewing	King Lear	820.42
Cade Ewing	Macbeth	797.5
Cade Ewing	Midsummer Night's Dream	935
Cade Ewing	Othello	870.83
Cade Ewing	Romeo and Juliet	733.33
Cade Ewing	Two Gentlemen of Verona	985.42
Callie Moon	Comedy of Errors	968.75
Callie Moon	Henry V	1687.5
Callie Moon	King Lear	1118.75
Callie Moon	Macbeth	1087.5
Callie Moon	Midsummer Night's Dream	1275
Callie Moon	Romeo and Juliet	1000
Callie Moon	Twelfth Night	1000
Chadwick Gardner	Henry V	1012.5
Chadwick Gardner	King Lear	671.25
Chadwick Gardner	Macbeth	652.5
Chadwick Gardner	Merchant of Venice	1046.25
Chadwick Gardner	Romeo and Juliet	600
Chadwick Gardner	Two Gentlemen of Verona	806.25
Clarke Stafford	Comedy of Errors	968.75
Clarke Stafford	Macbeth	1087.5
Clarke Stafford	Merchant of Venice	1743.75
Clarke Stafford	Midsummer Night's Dream	1275
Clarke Stafford	Much Ado about Nothing	681.25
Clarke Stafford	Othello	1187.5
Clarke Stafford	Two Gentlemen of Verona	1343.75
Dacey Mathis	Comedy of Errors	968.75
Dacey Mathis	Henry V	1687.5
Dacey Mathis	Macbeth	1087.5
Dacey Mathis	Merchant of Venice	1743.75
Dacey Mathis	Much Ado about Nothing	681.25
Dacey Mathis	Othello	1187.5
Dacey Mathis	Romeo and Juliet	1000
Dacey Mathis	Twelfth Night	1000
Darius Becker	Henry V	1237.5
Darius Becker	King Lear	820.42
Darius Becker	Macbeth	797.5
Darius Becker	Merchant of Venice	1278.75
Darius Becker	Midsummer Night's Dream	935
Darius Becker	Romeo and Juliet	733.33
Darius Becker	Twelfth Night	733.33
Darius Becker	Two Gentlemen of Verona	985.42

Query1_9		
ActorName	ShowName	Expr1002
Debra David	Comedy of Errors	710.42
Debra David	Henry V	1237.5
Debra David	King Lear	820.42
Debra David	Macbeth	797.5
Debra David	Merchant of Venice	1278.75
Debra David	Midsummer Night's Dream	935
Debra David	Romeo and Juliet	733.33
Denton French	Comedy of Errors	1937.5
Denton French	Henry V	3375
Denton French	King Lear	2237.5
Denton French	Macbeth	2175
Denton French	Merchant of Venice	3487.5
Denton French	Midsummer Night's Dream	2550
Denton French	Much Ado about Nothing	1362.5
Denton French	Romeo and Juliet	2000
Denton French	Twelfth Night	2000
Denton French	Two Gentlemen of Verona	2687.5
Donna Mullen	Comedy of Errors	1937.5
Donna Mullen	Henry V	3375
Donna Mullen	King Lear	2237.5
Donna Mullen	Macbeth	2175
Donna Mullen	Midsummer Night's Dream	2550
Donna Mullen	Much Ado about Nothing	1362.5
Donna Mullen	Othello	2375
Donna Mullen	Romeo and Juliet	2000
Donna Mullen	Two Gentlemen of Verona	2687.5
Ebony Sweeney	Henry V	1350
Ebony Sweeney	Macbeth	870
Ebony Sweeney	Midsummer Night's Dream	1020
Ebony Sweeney	Much Ado about Nothing	545
Ebony Sweeney	Romeo and Juliet	800
Elliott Saunders	Henry V	1687.5
Elliott Saunders	King Lear	1118.75
Elliott Saunders	Macbeth	1087.5
Elliott Saunders	Merchant of Venice	1743.75
Elliott Saunders	Midsummer Night's Dream	1275
Elliott Saunders	Much Ado about Nothing	681.25
Elliott Saunders	Othello	1187.5
Galvin Merritt	Comedy of Errors	968.75
Galvin Merritt	Henry V	1687.5
Galvin Merritt	King Lear	1118.75
Galvin Merritt	Macbeth	1087.5

Query1_9		
ActorName	ShowName	Expr1002
Galvin Merritt	Merchant of Venice	1743.75
Galvin Merritt	Midsummer Night's Dream	1275
Galvin Merritt	Othello	1187.5
Galvin Merritt	Two Gentlemen of Verona	1343.75
Garth York	Comedy of Errors	1162.5
Garth York	Henry V	2025
Garth York	Macbeth	1305
Garth York	Merchant of Venice	2092.5
Garth York	Midsummer Night's Dream	1530
Garth York	Much Ado about Nothing	817.5
Garth York	Othello	1425
Garth York	Twelfth Night	1200
Garth York	Two Gentlemen of Verona	1612.5
Hall Holt	Comedy of Errors	1162.5
Hall Holt	Macbeth	1305
Hall Holt	Midsummer Night's Dream	1530
Hall Holt	Othello	1425
Hall Holt	Romeo and Juliet	1200
Hall Holt	Twelfth Night	1200
Jared Larson	Henry V	1350
Jared Larson	King Lear	895
Jared Larson	Macbeth	870
Jared Larson	Merchant of Venice	1395
Jared Larson	Midsummer Night's Dream	1020
Jared Larson	Romeo and Juliet	800
Jared Larson	Two Gentlemen of Verona	1075
Jason Hyde	Comedy of Errors	581.25
Jason Hyde	Henry V	1012.5
Jason Hyde	Macbeth	652.5
Jason Hyde	Merchant of Venice	1046.25
Jason Hyde	Much Ado about Nothing	408.75
Jason Hyde	Twelfth Night	600
Leonard Hickman	Comedy of Errors	968.75
Leonard Hickman	Henry V	1687.5
Leonard Hickman	Macbeth	1087.5
Leonard Hickman	Much Ado about Nothing	681.25
Leonard Hickman	Romeo and Juliet	1000
Maggy House	Henry V	2475
Maggy House	King Lear	1640.83
Maggy House	Macbeth	1595
Maggy House	Much Ado about Nothing	999.17
Maggy House	Othello	1741.67

Query1_9		
ActorName	ShowName	Expr1002
Maggy House	Romeo and Juliet	1466.67
Maggy House	Twelfth Night	1466.67
Mary Rocha	Henry V	3375
Mary Rocha	King Lear	2237.5
Mary Rocha	Macbeth	2175
Mary Rocha	Merchant of Venice	3487.5
Mary Rocha	Midsummer Night's Dream	2550
Mary Rocha	Much Ado about Nothing	1362.5
Mary Rocha	Othello	2375
Mary Rocha	Romeo and Juliet	2000
Noelle Dillard	Henry V	2475
Noelle Dillard	King Lear	1640.83
Noelle Dillard	Macbeth	1595
Noelle Dillard	Midsummer Night's Dream	1870
Noelle Dillard	Much Ado about Nothing	999.17
Noelle Dillard	Othello	1741.67
Noelle Dillard	Romeo and Juliet	1466.67
Noelle Dillard	Two Gentlemen of Verona	1970.83
Odette Barker	Comedy of Errors	581.25
Odette Barker	Henry V	1012.5
Odette Barker	Macbeth	652.5
Odette Barker	Midsummer Night's Dream	765
Odette Barker	Much Ado about Nothing	408.75
Odette Barker	Romeo and Juliet	600
Odette Barker	Twelfth Night	600
Oprah Buckner	Henry V	2475
Oprah Buckner	King Lear	1640.83
Oprah Buckner	Macbeth	1595
Oprah Buckner	Midsummer Night's Dream	1870
Oprah Buckner	Much Ado about Nothing	999.17
Oprah Buckner	Romeo and Juliet	1466.67
Oprah Buckner	Twelfth Night	1466.67
Palmer Reyes	Henry V	1237.5
Palmer Reyes	Macbeth	797.5
Palmer Reyes	Merchant of Venice	1278.75
Palmer Reyes	Midsummer Night's Dream	935
Palmer Reyes	Much Ado about Nothing	499.58
Palmer Reyes	Twelfth Night	733.33
Palmer Reyes	Two Gentlemen of Verona	985.42
Prescott McMahon	Henry V	2025
Prescott McMahon	Macbeth	1305
Prescott McMahon	Merchant of Venice	2092.5

Query1_9		
ActorName	ShowName	Expr1002
Prescott McMahon	Much Ado about Nothing	817.5
Prescott McMahon	Romeo and Juliet	1200
Prescott McMahon	Twelfth Night	1200
Prescott McMahon	Two Gentlemen of Verona	1612.5
Summer Dickson	Comedy of Errors	710.42
Summer Dickson	Macbeth	797.5
Summer Dickson	Merchant of Venice	1278.75
Summer Dickson	Much Ado about Nothing	499.58
Summer Dickson	Othello	870.83
Summer Dickson	Romeo and Juliet	733.33
Veronica Decker	Comedy of Errors	387.5
Veronica Decker	Henry V	675
Veronica Decker	Macbeth	435
Veronica Decker	Much Ado about Nothing	272.5
Veronica Decker	Othello	475
Veronica Decker	Twelfth Night	400
Willow Woods	Comedy of Errors	968.75
Willow Woods	Henry V	1687.5
Willow Woods	Macbeth	1087.5
Willow Woods	Othello	1187.5
Willow Woods	Romeo and Juliet	1000
Zelda Fry	Henry V	1012.5
Zelda Fry	King Lear	671.25
Zelda Fry	Macbeth	652.5
Zelda Fry	Merchant of Venice	1046.25
Zelda Fry	Midsummer Night's Dream	765
Zelda Fry	Romeo and Juliet	600
Zelda Fry	Twelfth Night	600
Zelda Fry	Two Gentlemen of Verona	806.25

SECTION B OBJECT ORIENTED PROGRAMMING

FINAL OUTPUT:

All Bookings

A1	R200.0	Pd	
A2	R200.0	Pd	
A3	R200.0	Pd	
A5	R200.0	Pd	
A6	R200.0	Pd	
A7	R200.0		
A8	R200.0	Pd	
A9	R200.0	Pd	
A10	R200.0		
A11	R200.0	Pd	
A12	R200.0	Pd	
A13	R200.0	Pd	
A14	R200.0	Pd	
A15	R200.0		
B1	R140.0	Pd	Kolby Girls High School
B2	R140.0	Pd	Kolby Girls High School
B3	R140.0	Pd	Kolby Girls High School
B4	R140.0	Pd	Kolby Girls High School
B5	R140.0	Pd	Kolby Girls High School
B6	R140.0	Pd	Kolby Girls High School
B7	R140.0	Pd	Kolby Girls High School
B8	R140.0	Pd	Kolby Girls High School
B9	R140.0	Pd	Kolby Girls High School
B10	R140.0	Pd	Kolby Girls High School
B11	R140.0	Pd	Kolby Girls High School
B12	R140.0	Pd	Kolby Girls High School
C2	R200.0		
C3	R200.0		
C4	R200.0	Pd	
C5	R200.0		
C6	R200.0	Pd	
C7	R200.0	Pd	
C8	R200.0	Pd	
C10	R200.0		
C11	R200.0	Pd	
C14	R200.0		
D2	R140.0		Knysna Retirement Home
D3	R140.0		Knysna Retirement Home
D4	R140.0		Knysna Retirement Home
D5	R140.0		Knysna Retirement Home
D6	R140.0		Knysna Retirement Home
D7	R140.0		Knysna Retirement Home
D8	R140.0		Knysna Retirement Home
D9	R140.0		Knysna Retirement Home
D10	R140.0		Knysna Retirement Home
D11	R140.0		Knysna Retirement Home
D12	R140.0		Knysna Retirement Home
D13	R140.0		Knysna Retirement Home
D14	R140.0		Knysna Retirement Home
D15	R140.0		Knysna Retirement Home
E1	R160.0	Pd	
E3	R160.0	Pd	
E5	R160.0	Pd	
E6	R160.0	Pd	
E7	R160.0		
E8	R160.0		
E9	R160.0	Pd	
E10	R160.0		
E11	R160.0		
E12	R160.0	Pd	
E13	R160.0		
E14	R160.0		
F1	R160.0		
F2	R120.0	Pd	Witbank Hockey Club
F3	R120.0	Pd	Witbank Hockey Club
F4	R120.0	Pd	Witbank Hockey Club
F5	R120.0	Pd	Witbank Hockey Club
F6	R120.0	Pd	Witbank Hockey Club
F7	R120.0	Pd	Witbank Hockey Club
F8	R120.0	Pd	Witbank Hockey Club
F9	R120.0	Pd	Witbank Hockey Club
F10	R120.0	Pd	Witbank Hockey Club
F11	R160.0		
F12	R160.0	Pd	
F14	R160.0		
F15	R160.0	Pd	
G3	R100.0	Pd	Mellville Boy Scouts
G4	R100.0	Pd	Mellville Boy Scouts
G5	R100.0	Pd	Mellville Boy Scouts
G6	R100.0	Pd	Mellville Boy Scouts
G7	R100.0	Pd	Mellville Boy Scouts
G8	R100.0	Pd	Mellville Boy Scouts

G9	R100.0	Pd	Mellville Boy Scouts
G10	R100.0	Pd	Mellville Boy Scouts
G11	R100.0	Pd	Mellville Boy Scouts
G12	R100.0	Pd	Mellville Boy Scouts
G13	R100.0	Pd	Mellville Boy Scouts
G14	R100.0	Pd	Mellville Boy Scouts
G15	R100.0	Pd	Mellville Boy Scouts
H1	R160.0		
H2	R160.0		
H3	R160.0		
H5	R160.0		
H6	R160.0		
H7	R160.0		
H8	R160.0		
H9	R160.0		
H10	R160.0		
H11	R160.0		
H12	R160.0		
H13	R160.0		
H14	R160.0		
I1	R120.0		
I3	R120.0	Pd	
I4	R120.0		
I5	R120.0		
I6	R120.0	Pd	
I7	R120.0		
I9	R120.0	Pd	
I10	R120.0	Pd	
I11	R120.0	Pd	
I12	R120.0	Pd	
I13	R120.0	Pd	
J1	R60.0		Cape Town Drama Society
J2	R60.0		Cape Town Drama Society
J3	R60.0		Cape Town Drama Society
J4	R60.0		Cape Town Drama Society
J5	R60.0		Cape Town Drama Society
J6	R60.0		Cape Town Drama Society
J7	R60.0		Cape Town Drama Society
J8	R60.0		Cape Town Drama Society
J9	R60.0		Cape Town Drama Society
J10	R60.0		Cape Town Drama Society
J11	R60.0		Cape Town Drama Society
J12	R60.0		Cape Town Drama Society
J13	R60.0		Cape Town Drama Society
J14	R60.0		Cape Town Drama Society
K1	R60.0	Pd	Durban Gentleman's Club
K2	R60.0	Pd	Durban Gentleman's Club
K3	R60.0	Pd	Durban Gentleman's Club
K4	R60.0	Pd	Durban Gentleman's Club
K5	R60.0	Pd	Durban Gentleman's Club
K6	R60.0	Pd	Durban Gentleman's Club
K7	R60.0	Pd	Durban Gentleman's Club
K8	R60.0	Pd	Durban Gentleman's Club
K9	R60.0	Pd	Durban Gentleman's Club
K10	R60.0	Pd	Durban Gentleman's Club
K11	R60.0	Pd	Durban Gentleman's Club
K12	R60.0	Pd	Durban Gentleman's Club
K13	R60.0	Pd	Durban Gentleman's Club
K14	R60.0	Pd	Durban Gentleman's Club
K15	R60.0	Pd	Durban Gentleman's Club
L4	R120.0	Pd	
L6	R120.0	Pd	
L7	R120.0		
L8	R120.0		
L9	R120.0		

Group Bookings

Kolby Girls High School
Knysna Retirement Home
Witbank Hockey Club
Mellville Boy Scouts
Cape Town Drama Society
Durban Gentleman's Club

Outstanding Payments

Outstanding Payments : R 8760.0

Empty Seats

A4 B13 B14 B15 C1 C9 C12 C13 C15 D1 E2 E4 E15 F13 G1 G2 H4 H15 I2 I8 I14 I15 J15 L1 L2 L3 L5 L10
L11 L12 L13 L14 L15